Web Technologies MASE 2026 Project Specification 40% [Cross-modular with Continuous Build and Delivery]

1. Table of Contents

1. Table of Contents 1
2. Objective 2
3. Project Requirements 2
4. Project Architecture 4
5. Deadlines 5
6. Submissions 5
7. Marking Rubric 6
   1. **Written Report (35%)** 7
      Structure and Organisation (10%) 7
      Technical Content & Design (15%) 7
      Evaluation & Reflection (5%) 7
      Referencing & Academic Integrity (5%) 7
   2. **In-Person Demo & Viva (60%)** 8
      Live System Demonstration (25%) 8
      Technical Understanding (Viva) (30%) 8
      Professional Communication & Reasoning (5%) 9

2. Objective

The objective of this assignment is to design and develop a full-stack web application using **Spring Boot (REST API), (Frontend –JavaScript, jQuery, Bootstrap), and MySQL (Database)**. The project should showcase best practices in API development, entity relationships and error handling

You will:

- Design and implement **RESTful APIs** that follow best practices.
- Develop a **frontend** using Bootstrap and DataTables.
- Use **MySQL** as the relational database.
- Ensure **security, error handling, and scalability** in the implementation.
- Demonstrate proficiency in **modern UI features** such as DataTables, modals, glyphicons, and charts.

This project is part of a Cross-Modular Assessment with Continuous Build and Delivery. The pipeline will be assessed in the final in-person demo. The expectations and rubric for the pipeline part will be specified separately as part of the Continuous Build and Delivery module.

3. Project Requirements

User Stories & System Design

- Define **clear user stories** describing system functionalities.
- Between 5 and 10 User Stories is sufficient. The focus is on the implementation detail rather than the amount of functionality.

Backend - REST API (Spring Boot)

- Implement a **secure and scalable** API using **Spring Boot and Spring Data JPA**.

- Use **DTOs (Data Transfer Objects)** for structured data transfer.
- Implement **proper HTTP methods and status codes**.
- Implement **error handling and validation**.
- Use **JWT authentication and role-based access control**.

References

ONAP (Open Network Automation Platform)

See also the following for information on RESTful API Design Specification

https://wiki.onap.org/display/DW/RESTful+API+Design+Specification

Database - MySQL

- Implement a **structured relational database** using MySQL.
- Define **entity relationships** (One-to-Many or Many-to-Many).
- Use **Spring Data JPA** for database interaction.

Frontend – jQuery, JS, HTML, CSS, Bootstrap

Implement **Bootstrap UI elements**, including:

- **Cards**
- **Modals**
- **Glyphicons**
- **DataTables (if applicable)**
- **Charts/Graphs**

Documentation & Code Quality

- Provide a **README file** with setup instructions.
- Document **API endpoints** (Swagger or Postman collection).
- Follow **clean coding principles** (proper naming, comments, indentation).
- Use **Git for version control**.

Submission Requirements

1. **GitHub repository (link)** with the full project.
2. **API documentation** (Swagger/Postman collection).
3. **See requirements in Continuous Build and Delivery for CMA part.**

4. Project Architecture

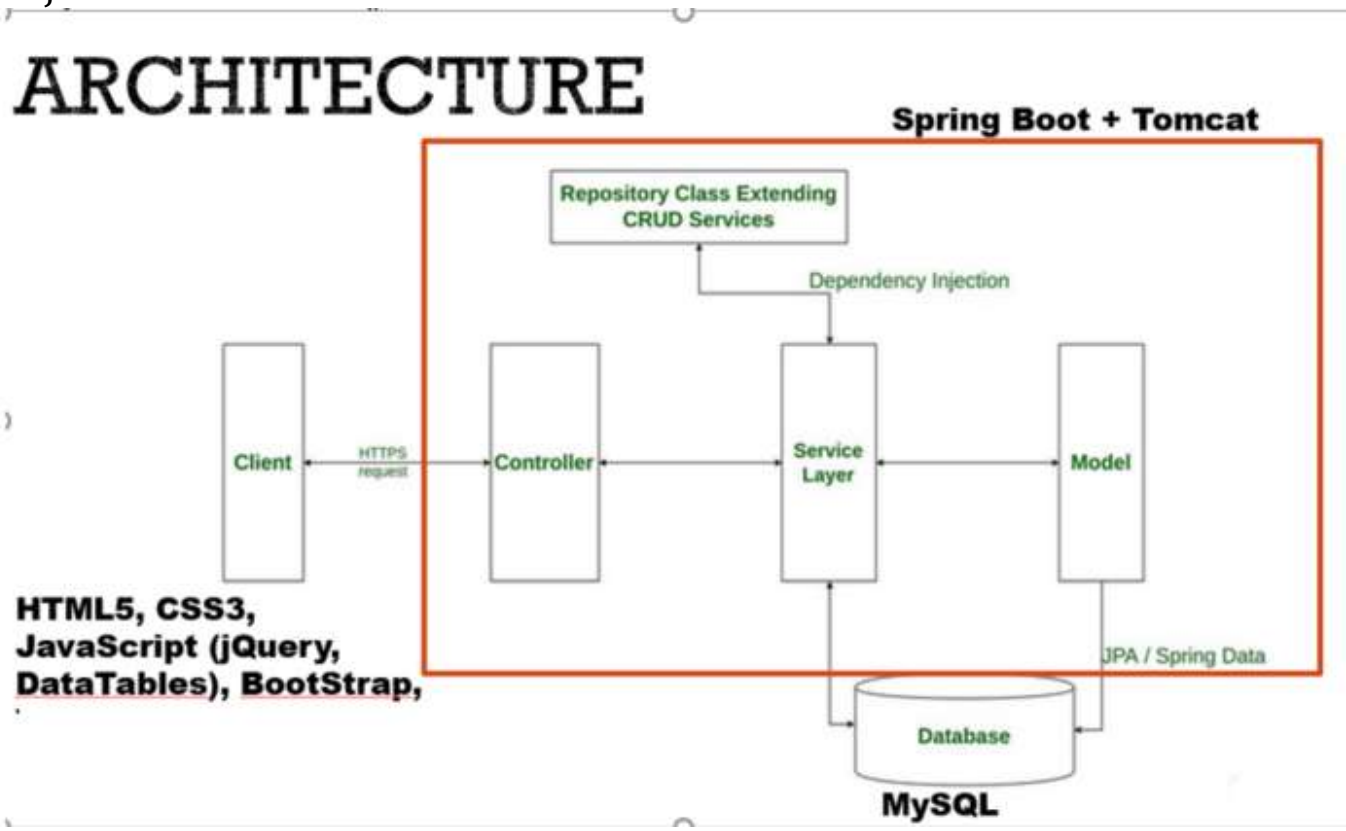

Figure 1 - Architecture

5. Deadlines

Submission#1 – List of User Stories with Acceptance Criteria, link to repo – wk5
Submit via a word document to moodle.
Submission#2 – Report and person demo/viva -wk10, & wk11

6. Submissions

	What to submit
Report	**Title Page** • Assignment Title: Full-Stack Web Application Development using Springboot, MySQL and JavaScript/jQuery frontend • Name, Student ID, Course Title, Date of Submission
	Table of Contents • A structured list of sections with page numbers
	Introduction • Overview of the project • Context and rationale
	System Design and User Stories • List of user stories with Acceptance Criteria (Given/When/Then) • Entity-Relationship Diagram (ERD) • UML Diagrams (if applicable)
	Backend – REST API Development Explanation of • How the API is secured • How DTO's are used (their benefit in this case) • Adherence to REST API Design Principles (What level according to Richardson Model) • Error Handling and Validation.
	Database Schema • MySQL schema

	• Entity relationships (One-to-Many or Many-to-many)
	Frontend Development • Adherence to SPA with JavaScript, jQuery,
	Conclusion & Future Work • Summary of key learnings. • Limitations and possible improvements • Scalability issues and options
	References • All sources of information, ideas, and data used in the assignment must be appropriately cited using the **Harvard referencing style**. This includes in-text citations and a full bibliography or reference list at the end of the report. Failure to adhere to this requirement will result in a deduction of marks or non-compliance with academic integrity policies.
	• Link to git repo
Presen tation	• Presentations during wk11/12 – 15 - 20 minutes. User Stories and Pipeline.

7. Marking Rubric

Component	**Weight**
Initial User Story List	5%
Written Report	35%
In-Person & Viva	60%
Total	100%

1. **Written Report (35%)**

Structure and Organisation (10%)

Excellent (70+)	Report is clearly structured, logically sequenced, and includes all required sections. Diagrams and explanations are well-integrated and referenced.
Good (55–69%)	Report is well organised with minor issues in flow or clarity. Most required sections present.
Satisfactory (40–54%)	Report includes required sections but lacks clarity, depth, or logical progression.
Fail (0–39%)	Poor structure, missing key sections, or difficult to follow.

Technical Content & Design (15%)

Excellent (70+)	Clear user stories with acceptance criteria. Strong system design (ERD/UML). REST API follows best practices (DTOs, HTTP semantics, JWT, validation). Database schema well-designed. Frontend uses Bootstrap components effectively.
Good (55–69%)	User stories and design present but lack depth. API largely correct with minor gaps. Database and frontend mostly correct.
Satisfactory (40–54%)	Partial or unclear user stories/design. API or database shows weaknesses. Frontend basic.
Fail (0–39%)	Major omissions or non-functional design elements.

Evaluation & Reflection (5%)

Excellent (70+)	Insightful evaluation of limitations, bugs, and architectural trade-offs with justified improvement suggestions.
Good (55–69%)	Reasonable evaluation with some reflection on limitations.
Satisfactory (40–54%)	Minimal evaluation; largely descriptive.
Fail (0–39%)	No meaningful evaluation.

Referencing & Academic Integrity (5%)

2. **In-Person Demo & Viva (60%)**

The session is a **live technical demonstration followed by examiner-led questioning**.

Duration: **~15/20 minutes per student**

Live System Demonstration (25%)

Excellent (70+)	System runs correctly. Backend, database, and frontend are demonstrated confidently. API behaviour, security, and UI features are clearly explained during use. Demo is User Story based.
Good (55–69%)	System mostly functional with minor issues. Explanation is clear but uneven. Not always clear what User Story is being shown.
Satisfactory (40–54%)	Partial demo; some components unclear or fragile. Not related to User Stories
Fail (0–39%)	System does not run or cannot be demonstrated meaningfully.

Technical Understanding (Viva) (30%)

Assessed through **direct questioning** by the examiners.

Example areas:

- Why specific architectural decisions were made
- How JWT authentication works in *this* system
- DTO purpose and trade-offs
- REST maturity (Richardson Model)
- Error handling and validation choices
- Database relationship decisions
- How validation is handled

Excellent (70+)	Answers show deep understanding, correct terminology, and ability to reason about trade-offs.
Good (55–69%)	Answers mostly correct with minor gaps or hesitations.
Satisfactory (40–54%)	Basic understanding but largely descriptive or memorised.
Fail (0–39%)	Unable to explain own system or decisions.

Professional Communication & Reasoning (5%)

Level	Descriptor
Excellent (70+)	Clear, confident explanations; engages with questions thoughtfully and professionally.
Good (55–69%)	Generally clear communication with minor issues.
Satisfactory (40–54%)	Communication lacks clarity or confidence.
Fail (0–39%)	Unable to explain or engage meaningfully.